Lecture #1: Sets, Permutations & Combinations

MATH 145 – Calculus I

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What is a Set?

A **set** is a collection of objects.

We often denote sets with capital letters: $A, B, C, \ldots.$

We say x is a **member** of a set A if x belongs to the collection of objects A, and we write $x \in A$. Otherwise, we write $x \notin A$.

What is a Set?

We usually write sets in **roster notation**:

$$A = \{1, 2, 3, 4, 5\}$$

$$B = \{x, y, z, w\}$$

The **empty set** is the set with no members.

$$\emptyset = \{\}$$

Set-Builder Notation

Sets can be specified using logical connectives and the membership relation using **set builder notation**.

$$C = \{x \mid x \in \mathbb{N} \land x \leq 3\} = \{0, 1, 2, 3\}$$

$$D = \{x \mid x > -2 \land x \leq 3\} = (-2, 3]$$

$$E = \{x \mid x > -5 \land x < 5\} = (-5, 5)$$

Subsets

$$D = \{x \mid x > -2 \wedge x \leq 3\} = (-2, 3]$$
$$E = \{x \mid x > -5 \wedge x < 5\} = (-5, 5)$$

Note that D is a **subset** of E, $D \subseteq E$, i.e. every element that is in D is also in E.

If every element in a set A is in B, but there are elements in B that are not in A, then we say that A is a **proper subset** of B, $A \subset B$.

Examples:

1. Is $\{1, 2, 4, 6, 9\}$ a subset of $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$?
2. Is the set of multiples of 7 a subset of the set of odd numbers?

Set Equality

Two sets A and B are equal (written $A = B$) when they have the same members (the **axiom of extensionality**).

$$A = B \iff (A \subseteq B) \wedge (B \subseteq A)$$

Russell's Paradox

These definitions are not perfect, and can cause some fun issues sometimes.

For example, Russell's Paradox: Let R be the set of all sets that are not members of themselves. Does R contain itself?

If R is not a member of itself, then its definition entails that it is a member of itself; yet, if it is a member of itself, then it is not a member of itself, since it is the set of all sets that are not members of themselves.

Some Important Sets

- **Natural Numbers:** $\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$.
- **Integers:** $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.
- **Rational Numbers:** $\mathbb{Q} = \left\{ \frac{a}{b} \,\middle|\, a, b \in \mathbb{Z}, b \neq 0 \right\}$.
- **Real Numbers:** $\mathbb{R}$.
- **Irrational Numbers:** $\mathbb{R} \setminus \mathbb{Q} = \{x \mid x \in \mathbb{R} \wedge x \notin \mathbb{Q}\}$.
- **Complex Numbers:** $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$.

Cardinality of a Set

The **cardinality** is the number of elemenets in a set, denoted as $|A|$.

$$|\{2,4,6,8\}| = 4$$

$$|\{3,6,9,12,15,18\}| = 6$$

$$|\mathbb{N}| = |\{0,1,2,3,4,\dots\}| = \infty$$

Careful, cardinality can be weird, there are many kinds of infinities. The two main ones are **countable** (like $\mathbb{Z}$) and **uncountable** (like $\mathbb{R}$).

Cardinality of a Set

Examples:

1. What is the cardinality of $\{1, 2, 3, 5, 8, 13\}$?
2. What is the cardinality of the set of all multiples of 3 less that 100?
3. Which set is larger: $\{1, 2, 3, 4\}$ or $\{1, 2, 3, 4, 5\}$?
4. Which set is larger: $\mathbb{N}$ or $\mathbb{R}$?
5. Which set is larger: $\mathbb{N}$ or $\mathbb{Z}$?
6. Which set is larger: $\mathbb{Z}$ or $\mathbb{Q}$?

Operations on Sets

New sets can be built up from existing sets (e.g., A and B) using logical connectives:

- The **union** of A and B, $A \cup B$, is the set of all elements in either A or B.
- The **intersection** of A and B, $A \cap B$, is the set of all elements in both A and B.

Operations on Sets

New sets can be built up from existing sets (e.g., A and B) using logical connectives:

- The **(set) difference)** or **complement** of A with B, $A \setminus B$, is the set of all elements in A not in B.
- The **complement of A** in the universe (if the context is clear) is the set of all elements in the universe not in A, denoted as $\overline{A}$ or A^C.
- A **universe** is a collection that contains all the entities one wishes to consider in a given situation. (Oftentimes, our universe will be something like $\mathbb{Z}$ or $\mathbb{R}$.)

Operations on Sets

Examples:

Let $A = \{2, 4, 6, 8\}$, $B = \{1, 3, 5, 7\}$ and $C = \{1, 2, 3, 4\}$.

1. What is the set $A \cup B$?
2. What is the set $A \cap C$?
3. What is the set $B \cup A$?
4. What is the set $C \setminus B$?
5. Prove that the set $(P \cap Q) \subseteq P$ for two sets P and Q.

De Morgan's Laws

De Morgan's Laws tell us how to combine unions and intersections with complements:

$$\overline{(A \cup B)} = \overline{A} \cap \overline{B}$$
$$\overline{(A \cap B)} = \overline{A} \cup \overline{B}$$

Break the line, change the sign!

De Morgan's Laws

These can be generalized as:

$$\overline{\left(\bigcup_{i\in I} A_i\right)} = \bigcap_{i\in I} \overline{A_i}$$
$$\overline{\left(\bigcap_{i\in I} A_i\right)} = \bigcup_{i\in I} \overline{A_i}$$

Break the line, change the sign!

The Product Rule/Multiplication Principle

The Product Rule:

For any finite sets A and B,

$$|A \times B| = |A| \cdot |B|,$$

where $A \times B = \{(a, b) \mid (a \in A) \wedge (b \in B)\}$.

Corollary: If each A_i is finite, then $\left|\prod_{i=1}^{k} A_i\right| = \prod_{i=1}^{k} |A_i|$.

Combinatorics

What does this tell us? Given a *sequence* of choices to make, the number of possible outcomes multiplies. The number can explode very quickly.

The key point is to conceptualize problems as a sequence of choices.

Our tools:

- **Exponentiation:**

$$a^b = \underbrace{a \times a \times a \times \cdots \times a}_{b \text{ times}}.$$

- **Factorial:**

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1.$$

Permutations

Question: How many ways are there to make an *ordered* subset of k items from a set of n total items?

"Ordered" means that being dealt the three of spades (3) and then the ten of hearts (10) is different than being dealt the ten of hearts (10) and then the three of spades (3).

Picking an ordered subset of size k from n items *with replacement* is easy, it's just n^k.

Picking an ordered subset of size k from n items *without replacement* is a bit more complicated, it is called a **k-permutation**.

k-Permutations

Picking an ordered subset of size k from n items *without replacement* is called a **k-permutation**.

The number of k–permutations of a set with n elements is denoted by $P(n,k)$, ${}_nP_k$ or $(n)_k$.

$$P(n,k) = \frac{n!}{(n-k)!}$$

k-Permutations Examples

Examples:

1. How many ways are there to place 5 students taken from a set of 8 students in a row of 5 chairs?
2. How many ways can I make a “leadership council” of students in this class, if the first person chosen is the “president”, the second person chosen is the “vice president” and the third person chosen is the “treasurer”?
3. How many *ordered* 5 card hands can be dealt from a standard 52-card deck?

Combinations

Question: What if order doesn't matter? How many ways are there to choose k items from a set of n total items *without replacement?

This is called a **combination**. This can donated as $C(n,k)$, ${}_nC_k$ or $\binom{n}{k}$ (called the **binomial coefficient**).

$$C(n,k) = \binom{n}{k} = \frac{n!}{(n-k)!\,k!}$$

Combinations

$$C(n,k) = \binom{n}{k} = \frac{n!}{(n-k)!\,k!}$$

Examples:

1. How many groups of 5 can I make from a class of 8 students?
2. How many ways can I make a “leadership council” of students in this class, if all three students are considered equal members?
3. How many 5 card poker hands can be dealt from a standard 52-card deck, assuming order doesn’t matter?

Other Types of Results

We can combine these formulas to solve more complex problems!

Example: How many ways are there to make a hand of 5 cards so that exactly two are spades and two are diamonds?

Example: How many ways are there to rearrange the letters in MISSISSIPPI?

We can use the **multinomial coefficient** to solve this problem:

$$\binom{n}{n_1 n_2 \ldots n_{k-1} n_k} = \frac{n!}{n_1!\, n_2!\, \ldots\, n_{k-1}!\, n_k!},$$

where $\sum_i n_i = n$.

Other Types of Results

We can combine these formulas to solve more complex problems!

Example: How many ways are there to choose 5 donuts from a selection of 3 *types* of donuts?

There are $C(n+r-1, n-1)$ ways to choose r things from n objects when repetitions of the objects are allowed!